Paper Reviews

1. Acoustic Anomaly Detection of Mechanical Failures in Noisy Real-Life Factory Environments

Yuki Tagawa, Rytis Maskeliūnas, Robertas Damaševičius (2021)

This paper develops a noise-tolerant acoustic anomaly-detection method for industrial machinery operating in realistic factory environments. It uses a generative adversarial network for sound-signal reconstruction and anomaly detection, with the MIMII dataset used for evaluation. The authors compare the approach with OC-SVM and an autoencoder–decoder baseline and investigate UKF preprocessing, MSE loss and L2 regularization. For our project, the paper directly supports the need for robust acoustic processing because factory noise can mask machine-fault signatures.

Relevant aspects: Acoustic anomaly detection; factory/background noise; MIMII; reconstruction-based detection; MSE; predictive maintenance.

Gap : It is acoustic-only and uses a more computationally expensive GAN-based approach, so it does not directly solve the ESP32-S3 multimodal TinyML problem.

https://www.mdpi.com/1282042

2. Rolling Element Bearing Diagnostics Using the Case Western Reserve University Data: A Benchmark Study

Wade A. Smith, Robert B. Randall

This paper systematically examines the Case Western Reserve University bearing dataset and applies established vibration-diagnostic techniques to create a benchmark for future algorithms. It shows that some records are easily diagnosed using envelope analysis while others are difficult or even undiagnosable, and that mechanical test-rig effects can sometimes influence the results more than the intended fault. For our project, it is important as the methodological reference for using CWRU as a vibration benchmark and for understanding the limitations of the dataset.

Relevant aspects: CWRU benchmark; vibration-based bearing diagnosis; envelope analysis; fault-frequency analysis; dataset limitations; benchmark evaluation.

Gap: It predates TinyML and multimodal sensing and focuses on conventional diagnostic methods rather than unsupervised edge anomaly detection.

https://www.researchgate.net/publication/276248554_Rolling_Element_Bearing_Diagnostics_Using_the_Case_Western_Reserve_University_Data_A_Benchmark_Study

3. TOYADMOS2: Another Dataset of Miniature-Machine Operating Sounds for Anomalous Sound Detection Under Domain Shift Conditions

Noboru Harada, Daisuke Niizumi, Daiki Takeuchi, Yasunori Ohishi, Masahiro Yasuda, Shoichiro Saito (2021)

ToyADMOS2 introduces a dataset designed to test anomalous sound detection under realistic domain shifts. The authors deliberately vary machine models and part configurations, operating speeds, microphone models and arrangements, and environmental noise. The dataset contains normal and anomalous machine sounds and is intended for unsupervised anomaly detection, where training primarily uses normal data. For our project, it is

especially relevant to cross-machine generalization and robustness: a model should not be evaluated only under the same machine, speed, sensor placement and environment used during training.

Relevant aspects: Unsupervised acoustic anomaly detection; domain shift; machine variation; operating-speed variation; microphone variation; environmental noise.

Gap: It is audio-only and does not address vibration fusion or microcontroller deployment.

https://dcase.community/documents/workshop2021/proceedings/DCASE2021Workshop_Harada_6.pdf

4. Signal Processing Optimization in Resource-Limited IoT for Fault Prediction in Rotating Machinery

Robertas Ūselis, Artūras Serackis, Raimondas Pomarnacki (2025)

This paper investigates fault prediction directly on an ultra-low-cost RP2040 microcontroller using vibration and motor-current signals. It extracts statistical time-domain features, applies PCA, and uses XGBoost models optimized through model distillation. The study explicitly evaluates memory and processing constraints and reports that distillation reduces model footprint while maintaining diagnostic performance. For our project, it strongly supports the need to design the signal-processing and ML pipeline around MCU constraints rather than simply selecting the most accurate desktop model. It is also useful for feature-selection and compression decisions.

Relevant aspects: Resource-constrained IoT; vibration; feature extraction; PCA; model compression/distillation; memory; edge deployment; predictive maintenance.

Gap : It uses supervised classification and motor current rather than acoustic–vibration healthy-only anomaly detection.

https://www.mdpi.com/3502198

5. A Multimodal TinyML-Based Predictive Maintenance Architecture for Industrial IoT in the 6G Era

Carlos Exequiel Garay et al. (2026)

This paper presents a multimodal TinyML predictive-maintenance architecture using vibration, acoustic and thermographic sensing with local inference on embedded devices. Its vibration pipeline evaluates anomaly-detection models on an actual Cortex-M4F target, including an INT8 fully connected autoencoder scored by reconstruction error. Acoustic inference uses log-Mel features, and the system also studies late multimodal fusion. The work additionally demonstrates physical testbed validation and per-installation healthy-baseline recalibration without retraining. For our project, this is the closest uploaded paper and is essential for accurately positioning our research contribution.

Relevant aspects: Multimodal sensing; acoustic + vibration; TinyML; INT8 autoencoder; reconstruction error; log-Mel features; edge inference; baseline recalibration; physical validation.

Gap : Because it already covers much of the broad concept, our novelty cannot simply be 'multimodal TinyML predictive maintenance.' It must be tied to our exact ESP32-S3/INMP441/MPU6050 implementation, fusion method, model and validation protocol.

https://www.mdpi.com/3986706

6. MIMII Dataset: Sound Dataset for Malfunctioning Industrial Machine Investigation and Inspection

Harsh Purohit, Ryo Tanabe, Kenji Ichige, Takashi Endo, Yuki Nikaido, Kaori Suefusa, Yohei Kawaguchi (2019)

MIMII introduces a public dataset of industrial machine sounds covering valves, pumps, fans and slide rails under normal and anomalous operating conditions. The intended task is unsupervised anomaly detection: normal sounds are used for training and anomalies are identified during testing. The dataset contains realistic anomalous conditions and background factory noise, with ten-second recordings collected using an eight-channel microphone array. For our project, MIMII is a foundational reference for the acoustic component, especially healthy-only autoencoder training, noise robustness and evaluation of industrial machine sound anomaly detection.

Relevant aspects: Industrial acoustic benchmark; normal-only training; anomalies; autoencoder baseline; factory noise; 10-second segments; multichannel recording.

Gap : It provides acoustic data only, so it cannot by itself validate the proposed vibration channel or multimodal fusion.

https://arxiv.org/pdf/1909.09347

7. IoT Device for Detecting Abnormal Vibrations in Motors Using TinyML

Stalin Arciniegas, Dulce Rivero, Jefferson Piñan, Elizabeth Diaz, Francklin Rivas (2025)

This paper presents a TinyML-based IoT device for detecting abnormal vibration in motor bearings. It combines spectral analysis and deep learning on resource-constrained edge hardware and reports real-time detection with low alert latency. The authors also discuss environmental noise and connectivity challenges and identify multimodal sensing as a future direction. For our project, it is a strong practical reference for the vibration side of the system: vibration acquisition, spectral/FFT-style feature extraction, motor-bearing anomaly detection, edge inference and the need to evaluate performance outside ideal laboratory conditions.

Relevant aspects: Vibration anomaly detection; motor/bearing faults; spectral analysis; TinyML; edge hardware; latency; environmental noise.

Gap: It does not implement the proposed acoustic channel, multimodal fusion or healthy-only reconstruction autoencoder.

https://link.springer.com/article/10.1007/s43926-025-00142-4

8. An Efficient Anomalous Sound Detection System for Microcontrollers

Yi-Cheng Lo, Tsung-Lin Tsai, Chieh-Wen Yang, An-Yeu Wu (2024)

This paper specifically addresses anomalous sound detection on microcontrollers. It identifies signal variation and limited MCU memory as major obstacles and proposes lightweight robust signal-processing techniques together with a memory-aware pruning algorithm. The resulting system is evaluated on the DCASE dataset with emphasis on both detection performance and resource efficiency. For our project, it is highly relevant to deploying acoustic anomaly detection on an ESP32-S3: it supports careful feature dimensionality, model-size reduction, pruning and hardware-aware optimization rather than designing a model that only works on a powerful computer.

Relevant aspects: Acoustic anomaly detection; microcontrollers; TinyML; lightweight processing; pruning; memory constraints; DCASE; edge intelligence.

Gap : It is acoustic-only and does not address the proposed vibration sensor or multimodal feature fusion.

https://www.mdpi.com/3055386

9. A Systematic Review of Anomaly and Fault Detection Using Machine Learning for Industrial Machinery

Syed Haseeb Haider Zaidi, Alex Shenfield, Hongwei Zhang, Augustine Ikpehai (2026)

This systematic review surveys machine-learning research on industrial anomaly and fault detection from 2010–2025. It categorizes supervised, unsupervised and hybrid approaches and discusses the pipeline from sensing and preprocessing through model deployment. The review highlights autoencoders and other deep-learning approaches for time-series and multimodal sensor data, while identifying continuing challenges such as data scarcity, class imbalance, limited generalization across equipment and scalable real-time deployment. For our project, it provides the broad state-of-the-art context needed to justify healthy-only anomaly detection, multimodal sensing, generalization and edge deployment.

Relevant aspects: State of the art; anomaly/fault detection; unsupervised learning; autoencoders; multimodal data; generalization; real-time deployment.

Gap: It is a review rather than an experimental implementation and therefore does not provide a specific ESP32-S3 architecture.

https://www.mdpi.com/3707642